# GVPT622 Lab Session

## Hypothesis Testing III

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# Set Up

- Load the packages we'll use today:

```
library(readxl)
library(tidyverse)
library(modelsummary)
library(infer)
```

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  - One-sample and two-sample $t$-tests

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# Introduction

- First: hypothesis tests for means and proportions
  - One-sample and two-sample $t$-tests
  - One-sample and two-sample proportion tests
- Then: the chi-squared test for a relationship between two categorical variables

# The Data

# Preparing the Data

```
cel <- read_excel("CELHouse93to118.xlsx", guess_max = 20000)

house118 <- cel |>
  rename(
    congress = "Congress number",
    name = "Legislator name, as given in THOMAS",
    dem = "1 = Democrat",
    female = "1 = female",
    vote_pct = "Percent vote received to enter this Congress",
    state_leg = "1 = served in state legislature",
    terms = "Seniority, number of terms served counting current"
  ) |>
  filter(congress == 118) |>
  mutate(party = if_else(dem == 1, "Democrat", "Republican"),
         party = factor(party, levels = c("Democrat", "Republican"))) |>
  select(name, party, female, vote_pct, state_leg, terms)
```

# The Data

```
house118
```

```
# A tibble: 448 x 6
   name             party      female vote_pct state_leg terms
   <chr>            <fct>       <dbl>    <dbl>     <dbl> <dbl>
 1 Adams, Alma      Democrat        1       63         1     6
 2 Aderholt, Robert Republican      0       84         0    14
 3 Aguilar, Pete    Democrat        0       58         0     5
 4 Alford, Mark     Republican      0       71         0     1
 5 Allen, Rick      Republican      0       60         0     5
# i 443 more rows
```

# One-Sample t-Test

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- A common rule of thumb: a member who wins with 60% or more holds a safe seat
- Is the average House member's vote share greater than 60%?
- $H_a$: $\mu > 60$
- $H_0$: $\mu = 60$

# The Sample Mean

```
mean(house118$vote_pct)
```

```
[1] 65.10714
```

# The t-Test

```
one_sample_t_test <- t.test(house118$vote_pct,
                            mu = 60,
                            alternative = "greater")

one_sample_t_test
```

```
        One Sample t-test

data:  house118$vote_pct
t = 9.6536, df = 447, p-value < 2.2e-16
alternative hypothesis: true mean is greater than 60
95 percent confidence interval:
 64.23514      Inf
sample estimates:
mean of x
 65.10714
```

- $p < 0.05$, so we reject $H_0$: the average member's vote share is greater than 60%

# Two-Sample t-Test

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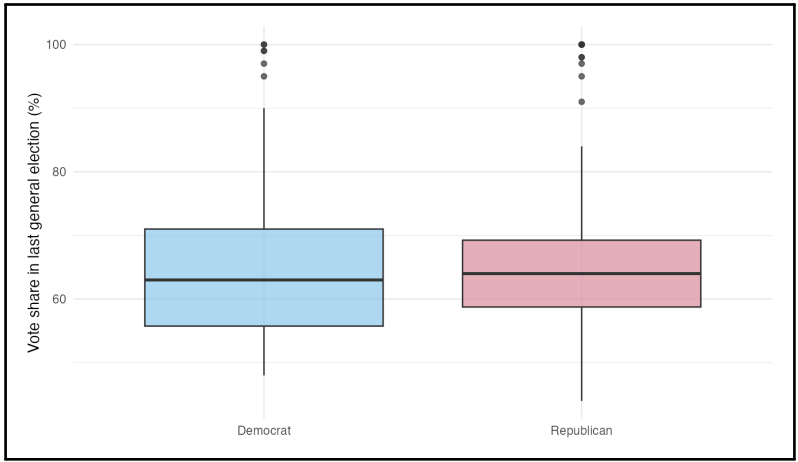
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- $H_0$: $\mu_{Dem} = \mu_{Rep}$

# Vote Share by Party

```
ggplot(house118, aes(x = party, y = vote_pct, fill = party)) +
  geom_boxplot(alpha = 0.7) +
  theme_minimal() +
  theme(legend.position = "none") +
  scale_fill_discrete(type = c("#8bc5ed", "#D88C9A")) +
  labs(x = NULL,
       y = "Vote share in last general election (%)")
```

# Vote Share by Party



# The t-Test

```r
two_sample_t_test <- t.test(vote_pct ~ party,
                            data = house118,
                            alternative = "greater",
                            var.equal = T)

two_sample_t_test
```

```
        Two Sample t-test

data:  vote_pct by party
t = -0.78094, df = 446, p-value = 0.7824
alternative hypothesis: true difference in means between group Democrat and
    group Republican is greater than 0
95 percent confidence interval:
 -2.571852       Inf
sample estimates:
  mean in group Democrat mean in group Republican
                64.68636                 65.51316
```

- Formula interface: DV $\sim$ IV. R subtracts the second group from the first (Democrat $-$ Republican), so "greater" matches $H_a$

# Reading the Output

- Formula interface: DV ~ IV. R subtracts the second group from the first (Democrat − Republican), so "greater" matches $H_a$
- $p = 0.78$, so we fail to reject $H_0$

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- The one-sample proportion test checks a prediction about a population proportion ($\pi$), for binary (0/1) outcomes
- Many House members get their start in a state legislature. Do fewer than half come from one?
- $H_a$: $\pi < 0.5$
- $H_0$: $\pi = 0.5$

```
x_successes <- sum(house118$state_leg)
n_obs <- nrow(house118)

x_successes
n_obs
```

```
[1] 207
[1] 448
```

# The Proportion Test

```r
prop_test_one_sample <- prop.test(x = x_successes,
                                  n = n_obs,
                                  p = 0.5,
                                  alternative = "less")

prop_test_one_sample
```

```
        1-sample proportions test with continuity correction

data:  x_successes out of n_obs, null probability 0.5
X-squared = 2.4308, df = 1, p-value = 0.05949
alternative hypothesis: true p is less than 0.5
95 percent confidence interval:
 0.0000000 0.5020253
sample estimates:
        p
0.4620536
```

- 46.2% served in a state legislature, but $p = 0.059$

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- At $\alpha = 0.05$, we fail to reject $H_0$

# Two-Sample Proportion Test

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- $H_0$: $\pi_{Dem} = \pi_{Rep}$

# Successes and Trials by Party

```
women_by_party <- house118 |>
  group_by(party) |>
  summarise(successes = sum(female),
            n = n())

women_by_party
```

```
# A tibble: 2 x 3
  party      successes     n
  <fct>          <dbl> <int>
1 Democrat          95   220
2 Republican        36   228
```

# The Proportion Test

```r
prop_test_two_sample <- prop.test(x = women_by_party$successes,
                                  n = women_by_party$n,
                                  alternative = "greater")

prop_test_two_sample
```

```
        2-sample test for equality of proportions with continuity correction

data:  women_by_party$successes out of women_by_party$n
X-squared = 39.29, df = 1, p-value = 1.826e-10
alternative hypothesis: greater
95 percent confidence interval:
 0.2016704 1.0000000
sample estimates:
   prop 1    prop 2
0.4318182 0.1578947
```

# Reading the Output

- `x` and `n` are vectors, one entry per group (Democrat first)

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- `x` and `n` are vectors, one entry per group (Democrat first)
- 43.2% vs. 15.8%, $p \approx 2 \times 10^{-10}$. We eject $H_0$

# Chi-Squared Test for Independence

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## Safe Seats and Seniority

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- Is a member's tenure associated with how safe their seat is?
  - Seat: Competitive ($<$55%), Leaning (55–65%), Safe (65%+)
  - Tenure: Freshman, 2–4 terms, 5+ terms
- $H_a$: seat safety and tenure are associated
- $H_0$: seat safety and tenure are independent

# Creating the Categories

```
house118 <- house118 |>
  mutate(
    seat = case_when(vote_pct < 55 ~ "Competitive (<55%)",
                     vote_pct < 65 ~ "Leaning (55-65%)",
                     TRUE ~ "Safe (65%+)"),
    seat = factor(seat, levels = c("Competitive (<55%)", "Leaning
    (55-65%)",
                                   "Safe (65%+)")),
    tenure = case_when(terms == 1 ~ "Freshman",
                       terms <= 4 ~ "2-4 terms",
                       TRUE ~ "5+ terms"),
    tenure = factor(tenure, levels = c("Freshman", "2-4 terms", "5+
    terms"))
  )
```

# Observed Counts

```
datasummary_crosstab(seat ~ tenure, data = house118,
                     statistic = 1 ~ 1 + N + Percent("col"),
                     output = "markdown")
```

# Observed Counts

| seat | | Freshman | 2-4 terms | 5+ terms | All |
|---|---|---|---|---|---|
| Competitive (<55%) | N | 33 | 34 | 9 | 76 |
| | % col | 39.8 | 19.8 | 4.7 | 17.0 |
| Leaning (55-65%) | N | 24 | 67 | 78 | 169 |
| | % col | 28.9 | 39.0 | 40.4 | 37.7 |
| Safe (65%+) | N | 26 | 71 | 106 | 203 |
| | % col | 31.3 | 41.3 | 54.9 | 45.3 |
| All | N | 83 | 172 | 193 | 448 |
| | % col | 100.0 | 100.0 | 100.0 | 100.0 |

# Expected Counts

- If $H_0$ is true, the counts in each cell would reflect the overall row and column totals

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- If $H_0$ is true, the counts in each cell would reflect the overall row and column totals
- The expected count for any cell:

$$\text{Expected Count} = \frac{\text{Row Total} \times \text{Column Total}}{\text{Grand Total}}$$

- $\chi^2$ measures the total difference between the observed counts and the counts expected under $H_0$:

$$\chi^2 = \sum \frac{(\text{Observed} - \text{Expected})^2}{\text{Expected}}$$

```
chi_sq <- chisq.test(house118$seat, house118$tenure)

chi_sq
```

```
        Pearson's Chi-squared test

data:  house118$seat and house118$tenure
X-squared = 53.715, df = 4, p-value = 6.038e-11
```

- The p-value is extremely small, so we reject $H_0$: Seat safety and tenure are associated

# Conclusion

- $t$-tests for comparing means of continuous variables

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- $t$-tests for comparing means of continuous variables
- Proportion tests for comparing proportions of binary variables
- Chi-squared tests for the association between two categorical variables
- The core logic is the same for all of them: assume $H_0$ is true, calculate a test statistic, and see how likely that statistic is under the null distribution

# Citation

Center for Effective Lawmaking. 2025. "House Legislative Effectiveness Scores, 93rd–118th Congresses." `https://thelawmakers.org`.

Volden, Craig, and Alan E. Wiseman. 2014. *Legislative Effectiveness in the United States Congress: The Lawmakers*. New York: Cambridge University Press.

# Questions?